



HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering

CSE221- Computer Algorithms

Batch: 25th

Date: 18-06-2025

Mid-Term Examination

Semester: Spring 2025

Time: 1 Hour 30 Minutes

Total Marks: 30

Answer any five questions of the followings.

5 × 6 = 30

CLOs Marks

1. (a) Explain the types of algorithm efficiencies. CLO1 2
(b) Define asymptotic notation. Describe the steps involved in the algorithm analysis framework. CLO1 4
2. (a) Write the properties of Max-Heap. CLO3 2
(b) Analyze the complexity from the following recurrence: CLO1 4

$$T(n) = \begin{cases} 1 & \text{if } n \leq 1 \\ 2 \times T\left(\frac{n}{2}\right) + n & \text{otherwise} \end{cases}$$
3. (a) Define feasible solution and optimal solution. CLO1 2
(b) Suppose you have 10 elements in an array as follows: CLO2 4

Address	1	2	3	4	5	6	7	8	9	10
Elements	5	-8	35	11	85	-17	25	31	2	12

Draw the recursion tree to find the maximum and minimum elements from the array using recursive Min-Max Algorithm.
4. (a) Explain the core process of Divide & Conquer problem-solving approach. CLO2 2
(b) Construct the 'Merge' function to use in Merge Sort Algorithm. CLO5 4
5. (a) Explain the components of Greedy Method. CLO5 2
(b) Suppose you have four slots and six jobs with the following profits and deadlines respectively: CLO5 4
 $\{p_1, p_2, p_3, p_4, p_5, p_6\} = \{20, 17, 12, 10, 9, 8\}$ and
 $\{d_1, d_2, d_3, d_4, d_5, d_6\} = \{3, 1, 4, 1, 3, 2\}$
 If each job takes one unit of time to execute, find the optimal solution using Greedy Algorithm.
6. (a) What is the role of sorting in greedy algorithms? Why is it often necessary? CLO5 3
(b) Consider the following instance of knapsack problem: CLO5 3
 Number of objects, $n = 6$;
 Capacity of knapsack, $m = 17$;
 Profits, $\{p_1, p_2, p_3, p_4, p_5, p_6\} = \{10, 5, 18, 17, 21, 18\}$ and
 Weights, $\{w_1, w_2, w_3, w_4, w_5, w_6\} = \{3, 2, 4, 2, 4, 5\}$.
 Calculate the optimal solution using Greedy Method.



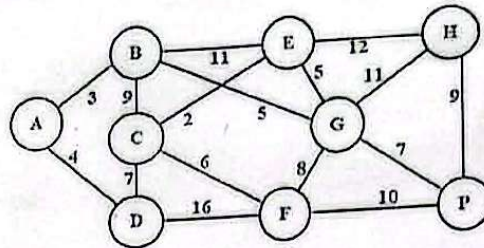
7. (a) Differentiate between Prim's Algorithm and Kruskal's Algorithm.
(b) Apply Kruskal's Algorithm to generate the minimum spanning tree for the following graph:

CLO3

2

CLO3

4





HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering

CSE225: Digital Logic Design

Batch: 25th
Date: 30/06/2025

Midterm Examination

Semester: Spring 2025
Time: 1 Hour 30 minutes
Total Marks: 30

Answer any five questions of the following.

$5 \times 6 = 30$

1. (a) Find the 9's and 10's complements of the number $(87283)_{10}$ CLO3 2
(b) Perform the following operations using complements and show all steps clearly. CLO3 4
 - i. $(2412)_5 - (3341)_5$ using 4's complement
 - ii. $(101101.101)_2 - (110011.11)_2$ using 2's complement
2. (a) Design a 3-bit even parity generator circuit where the output is parity bit (P) such that total number of 1's (including P) is even. CLO1 4
 - i. Use K-map to minimize the expression.
 - ii. Calculate the cost of the circuit.
- (b) Prove the following expression using Boolean algebra: CLO2 2
$$A(A' + B)(A' + B') = 0$$
3. (a) Simplify the boolean expression using K-map. CLO2 3
$$F = (A + B'C + D')(A'B + C'D)(A + C + D) + A'B'C'D$$
- (b) Reduce the following function to exactly 6 literals. CLO2 3
$$F(W, X, Y, Z) = W'X'Y + WXY' + W'XZ + WX'Z'$$
4. (a) Given three Boolean functions: CLO2 3
$$F_1 = AB' + A'B$$
$$F_2 = (A + B')(A' + B)$$
$$F_3 = A \oplus B$$

Prove that $F_1 = F_2 = F_3$ using Boolean algebra laws.
- (b) Explain with an example why a prime implicant might not be essential. Create a specific 3-variable function that demonstrates this concept. CLO3 3

5. A digital voting system for 4 members (A,B,C,D) requires a majority vote to pass a proposal. However, member A has veto power (can block any proposal regardless of others' votes). CLO5 6
- Create the truth table
 - Find both simplified SOP and POS forms
 - Compare the cost between SOP and POS implementations.
6. For the function $F(A,B,C,D) = \sum m(0,2,5,6,7,8,10,13,14,15)$ CLO3 6
- Identify ALL prime implicants
 - Identify ALL essential prime implicants
 - If the above function had don't care conditions at $m(1,4,12)$, how would the prime implicants and essential prime implicants change?
7. A smart elevator system for a 3-floor building has the following logic: CLO5 6
- Floor buttons: F_1, F_2, F_3 (1 when pressed)
 - Current floor sensors: C_1, C_2 (00=floor 1, 01=floor 2, 10=floor 3)
 - The elevator motor (M) should run only when:
Any floor button is pressed AND it's not the current floor.
- * Design the Boolean function for motor (M) using K-map simplification.

Handwritten notes and diagrams, including a K-map grid and some text, are visible in the bottom right corner of the page.



HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology Department of Computer Science and Engineering CSE 223: Object Oriented Programming

Batch: 25th

Semester: Spring 2025

Time: 1 Hour 30 Minutes

Date: 23/06/2025

Mid-Term Examination

Total Marks: 30

Answer any five questions of the followings.

5 × 6 = 30

	CLOs	Marks
1(a). Define the model of Object Oriented Programming (OOP).	CLO 1	1
1(b). Write the significant properties of OOP models.	CLO 1	2
1(c). Briefly explain the class and interaction models of OOP.	CLO 2	3
2(a). Write the differences between Object Oriented Programming and Structured Programming languages.	CLO 1	2
2(b). Explain the basic characteristics of OOP.	CLO 1	4
3(a). Briefly explain the uses of new and final keywords in OOP.	CLO 3	2
3(b). Explain the relationship among Java Development Kit, Java Runtime Environment, and Java Virtual Machine.	CLO 3	4
4(a). Illustrate the basic exception-class hierarchy, with Throwable at the top. Include at least two subclasses of Error, Exception, and RuntimeException.	CLO 3	2
4(b). Consider a class named Account with the following information: i. double balance - the balance of the account ii. void debit (double amount) - this method subtracts the amount from the balance iii. void credit (double amount) - this method adds the amount directly to the balance The amount provided in the debit and credit method can't be negative. The balance of the account can't also be negative. Write Java code for the following: i. A custom exception named InvalidAmountException that triggers when the given amount is negative and shows the message 'The given amount can't be negative'. ii. A custom exception named InvalidBalanceException that triggers when balance goes negative and shows the message 'The account balance can't be less than zero'. iii. The Account class to trigger these exceptions when needed.	CLO 5	4
5(a). What is the output when the following code fragment is executed?	CLO 3	1

```
int found = 0, count = 5;
if (!found || -count == 0)
    System.out.println( "danger" );
System.out.println( "count = " + count );
```

- 5(b) The Stanton measure of an array is computed as follows. Count the number of 1s in the array. Let this count be n . The Stanton measure is the number of times that n appears in the array. For example, the Stanton measure of {1,4,3,2,1,2,3,2} is 3 because 1 occurs 2 times in the array and 2 occurs 3 times. Write a method named `stantonMeasure` that returns the Stanton measure of its array argument. The function prototype is

int stantonMeasure(int[] a)

Examples:

if a is	return	reason
{1}	1	1 occurs 1 time, 1 occurs 1 time
{0}	1	1 occurs 0 times, 0 occurs 1 time
{3,1,1,4}	0	1 occurs 2 times, 2 occurs 0 times
{1,3,1,1,3,3,2,3,3,4}	6	1 occurs 3 times, 3 occurs 6 times
{}	0	1 occurs 0 times, 0 occurs 0 times

- 6(a). Briefly explain method overloading with example. CLO 1 3
- 6(b). Write a Java program to create a class called "Book" with instance variables title, author, and price. Implement a default constructor and two parameterized constructors:
- One constructor takes title and author as parameters.
 - The other constructor takes title, author, and price as parameters.
 - Print the values of the variables for each constructor.
- 7(a). Briefly explain access modifier in Java. CLO 1 2
- 7(b). Consider the following code segment. There are 4 (four) method calls in the main method. How many of them will generate compiler error and why? CLO 3 4

```
public class VarArgsTest {
    static void f(int... v) {}
    static void f(String msg, boolean ... v) {}
    static void f(String msg, int... v) {}
    static void f(int n, int ... v) {}
    static void f(double ... v) {}
    public static void main(String[] args) {
        f("int", 10, 20, 30);
        f("boolean", true, false, false);
        f();
        f(1, 2, 3);
    }
}
```




HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE229: Computer Architecture

Batch: 25th
Date: 28/06/2025

Midterm Examination

Semester: Spring 2025
Time: 1 Hour 30 minutes
Total Marks: 30

Answer any five questions of the following.

5 × 6 = 30

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|--------|--|------|---|
| 1. (a) | Distinguish between Computer Architecture and Computer Organization with examples. | CLO1 | 3 |
| (b) | Draw the hierarchical structure of a computer system showing Structure and Function at each level. | CLO1 | 3 |
| 2. (a) | Why do computers use 2's complement instead of sign-magnitude? Demonstrate with a binary addition example. | CLO4 | 2 |
| (b) | State overflow rule and subtraction rule with one example for each. | CLO4 | 2 |
| (c) | Design a hardware adder/subtractor block diagram with overflow detection for 2's complement numbers. | CLO3 | 2 |
| 3. (a) | Simulate the multiplication operation 1001×1100 showing register states (A, Q, M, C) at each step using shift-and-add method. | CLO4 | 4 |
| (b) | Explain Booth's algorithm principle and its advantages over standard multiplication. | CLO5 | 2 |
| 4. (a) | Simulate $(-5) \times (3)$ using Booth's algorithm with 4-bit representation. Show complete register traces. | CLO4 | 3 |
| (b) | Perform division operation $1000 \div 0101$ using the restoring division algorithm. Show all steps including shifts and restores. | CLO4 | 3 |
| 5. (a) | Draw flowcharts for Floating-point division operation. | CLO3 | 2 |
| (b) | Draw the flowchart for floating-point addition/subtraction showing normalization steps. | CLO3 | 4 |

6. (a) Draw the complete instruction cycle with interrupts state diagram showing all possible states and transitions. CLO3 2
- (b) Consider a hypothetical processor with the following memory contents (in hexadecimal): CLO2 4

Address	Contents
400	1A50
401	5A51
402	2A51
...	...
A50	0001
A51	000B
A52	0000

Trace through the execution of the program starting from PC=400, showing the contents of PC, IR, and AC after each step. Create table similar to the one shown in the reference material.

7. (a) Draw the complete computer system diagram showing CPU registers (PC, IR, MAR, MBR, I/O AR, I/O BR), memory, and I/O modules connected via system bus. CLO3 3
- (b) A computer system needs to execute the following program: CLO3 3
- LOAD R1, A ; Load value A into register R1
- ADD R1, B ; Add value B to R1
- STORE R1, C ; Store result in C
- HALT ; Stop execution
- Trace through the fetch-execute cycle for the first instruction (LOAD R1, A), showing the data flow and control signals at each step.

Faculty of Science, Engineering and Technology Department of Computer Science and Engineering CSE 227: Data Communication

Batch: 25th

Date: 25/06/2025

Mid-Term Examination

Semester: Spring 2025

Time: 1 Hour 30 Minutes

Total Marks : 30

Answer all the questions:

5 × 6 = 30

	CLOs	Marks
1. (a) Explain the five components of a data communication system.	CLO1	3
(b) What are the differences between simplex and half-duplex transmission?	CLO1	3
2. (a) Show the encapsulation and decapsulation process in TCP/IP model.	CLO5	2
(b) Write the differences between the duties of the IETF and IRTF.	CLO4	4
3. (a) Write the reasons for lack of OSI Model's success.	CLO4	3
(b) Explain Multiplexing and De-multiplexing in TCP/IP protocol suite.	CLO4	3
4. (a) A file contains 2 million bytes. How long does it take to download this file using a 56-Kbps channel and 1-Mbps channel?	CLO1	3
(b) Describe Nyquist Bit Rate and Shannon Capacity.	CLO1	3
5. (a) A nonperiodic composite signal contains frequencies from 10 to 30 KHz. The peak amplitude is 10 V for the lowest and the highest signals and is 30 V for the 20-KHz signal. Assuming that the amplitudes change gradually from the minimum to the maximum, draw the frequency spectrum.	CLO3	3
(b) Explain Amplitude, Frequency and Phase.	CLO3	3
6. (a) Briefly explain Data rate limits.	CLO1	3
(b) Name and explain three types of transmission impairment.	CLO3	3
7. Define the followings:	CLO5	6
I. Throughput		
II. Latency		
III. Bandwidth Delay Product		

HAMDARD UNIVERSITY BANGLADESH
Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
ENS221: Environmental Science

Midterm Examination

Date: 21/06/25

Semester: Spring 2025

Time: 1.5 Hours

Total Marks: 30

Part A (Answer any Two)

		CLOs	Marks
1.	a) Define environmental science.	CLO1	02
	b) Environmental science is a multi-disciplinary approach'. Justify it in your own words.	CLO1	03
	c) Apply your knowledge of environmental science to identify and explain the major threats on environment. Support your answer with real-world examples.		05
2.	a) Apply your understanding of environmental issues to explain why protecting the environment is critically important in today's world.	CLO1	06
	b) Apply your understanding of eutrophication to explain the excessive use of fertilizers in agriculture and cleaning materials of households can lead to water pollution and affect aquatic life.	CLO1	04
3.	a) 'Global warming is a part of climate change'. Justify it.	CLO1	02
	b) Analyze the greenhouse mechanism by breaking down its key components and processes. How do human activities intensify this natural process, and what are the resulting impacts on global climate systems?		03
	c) Differentiate between point source and non-point source pollutant. Provide one example of each and explain how each type of pollution affects the environment.	CLO1	05

Part B (Answer any One)

4.	a) The ozone layer acts as the blanket of the atmosphere. Justify it.	CLO2	02
	b) Analyze the key features of troposphere and mesosphere.	CLO2	04
	c) Compare and analyze the characteristics and functions of the ionosphere and exosphere. How do their roles differ in Earth's atmospheric system?	CLO2	04
5.	a) Delineate coastal area with figure.	CLO2	01
	b) Analyze its key principles of Climate Smart Agriculture (CSA) and explain why the CSA is important for food security and the environment.	CLO2	04
	c) Apply your understanding of climate change to explain the mechanism of sea level rise. Based on this understanding, identify and explain at least three specific effects that sea level rise can have on coastal areas.	CLO2	05